BENJAMIN NICHOLSON

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Education

Stony Brook University Stony Brook, NY
MS in Applied Mathematics and Statistics; GPA: 3.7/4.0 *Aug 2025 – Dec 2026*

- Relevant Coursework: Stochastic Calculus, Advanced Stochastic Models, Statistical Methods in Finance

Seton Hill University Pittsburgh, PA
BS in Data Science; GPA: 3.7/4.0 *Jan 2022 – May 2025*

- Relevant Coursework: Machine Learning, Numerical Analysis, Mathematical Modeling, Graph Theory
- Honors & Leadership: NCAA DII Soccer Team Athletic Scholarship & Captain, 2025 Data Science Honors Award

Work Experience

Siemens - Foundational Technologies (RPD) Princeton, NJ
Foundational Time Series Modeling Intern *May 2026 – Aug 2026*

- Built rolling multi-step electricity price forecasting pipeline for 5-minute and 1-hour electricity markets, integrated into smart battery charging service and arbitrage trading strategy
- Formulated Mixed-Integer Linear Programming (MILP) framework to optimize arbitrage revenue in real time markets while mitigating battery degradation
- Backtested few-shot Time Series Foundation Model which utilized key-query embedded historical retrieval mechanism outperforming baseline AutoARIMA, XGBoost & LSTM by 45%
- Built XGBoost classification model incorporating graph topology network flow to model physical constraints of grid, mitigating extreme tail risk in regression forecasts and achieving a recall of 95% on price spikes

Glimm Analytics New York, NY
Quantitative Research Intern *Feb 2026 – May 2026*

- Engineered Double Deep Q-Network regime-switching trading agent in US large-cap equities with backtested out of sample Sharpe ratio of 1.3 as well as outperforming SPY by 21% in annualized returns
- Developed ensemble GARCH-LSTM model on intraday volatility; reduced mean absolute error by 54% vs GARCH which served as a market state-risk feature for the trading agent

EY-Parthenon London, UK
Winter AI Intern *Jan 2026*

- Designed multi-agent LLM with AI trace retrieval reducing computational load on RLHF/SFT

Projects

Multi-Agent Reinforcement Learning Calibration of Options Stony Brook University
Replicated Vadori 2022 Paper from JP Morgan's Deep Hedging *Mar 2026 – May 2026*

- Formulated option price calibration as a cooperative multi-agent stochastic game where 10^5 Monte Carlo trajectories act as individual agents, collectively replicating market volatility surfaces, achieving a 70% reduction in MAE over Black Scholes baseline surfaces
- Utilized Actor-Critic architecture using PPO with price-based reward function and vega-weighting to enhance numerical stability and accuracy for deep OTM/ITM options.

Cross-Currency Dynamics in Crypto Market Stony Brook University
Joint Paper from IAQF 2026 Competition *Jan 2026 – Feb 2026*

- Analyzed breakdown in USDC arbitrage efficiency with 80GB+ tick level L2 microstructure data. Identified structural shift in time series statistical properties through Binary Segmentation and AR(1) half life of price deviations
- Developed arbitrage indicator function which highlighted 75% of depegging period having arbitrage trading opportunities vs 2% in normal period across multiple exchanges and trading sizes while incorporating slippage from walking the book and trading costs derived from exchange specific costs

Skills

Technical Skills: Deep Reinforcement Learning, Machine Learning (Random Forests, XGBoost), Monte Carlo Simulations, Time Series Analysis (Statistical, Deep Learning & Transformer), Differential & Information Geometry
Programming Skills: Python, C++, Git, SQL, R
Packages/Libraries: NumPy, Pandas, Polars, PyTorch, Plotly, SciPy, SimPy

Competitions

EY Data Challenge 2025: Ranked Top 1% globally (10K + particpants) by building ML temperature anomaly detections using Random Forest with Bayesian Optimization (R2: 0.9606)